

Q1 2021 (01 Sep Shift 2)

The half life period of radioactive element x is same as the mean life time of another radioactive element y .

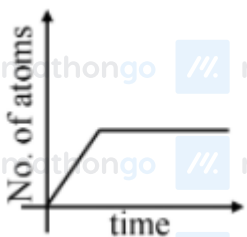
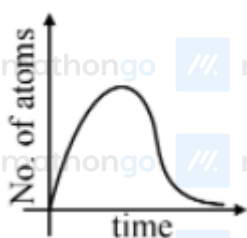
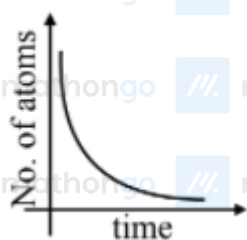
Initially they have the same number of atoms. Then :

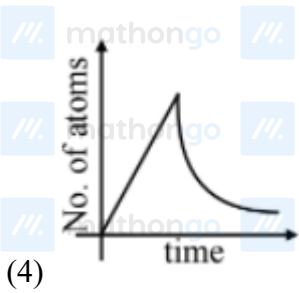
- (1) x -will decay faster than y .
- (2) y - will decay faster than x .
- (3) x and y have same decay rate initially and later on different decay rate.
- (4) x and y decay at the same rate always.

Q2 2021 (31 Aug Shift 1)

A sample of a radioactive nucleus A disintegrates to another radioactive nucleus B , which in turn disintegrates to some other stable nucleus C . Plot of a graph showing the variation of number of atoms of nucleus B versus time is :

(Assume that at $t = 0$, there are no B atoms in the sample)





Q3 2021 (27 Aug Shift 1)

There are 10^{10} radioactive nuclei in a given radioactive element, Its half-life time is 1 minute. How many nuclei will remain after 30 seconds ? ($\sqrt{2} = 1.414$)

(1) 2×10^{10}

(2) 7×10^9

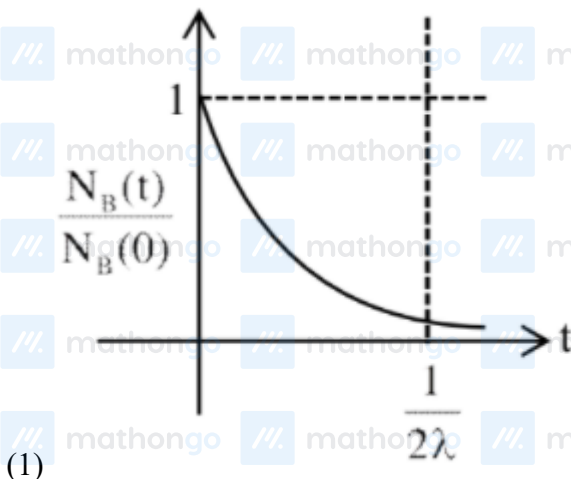
(3) 10^5

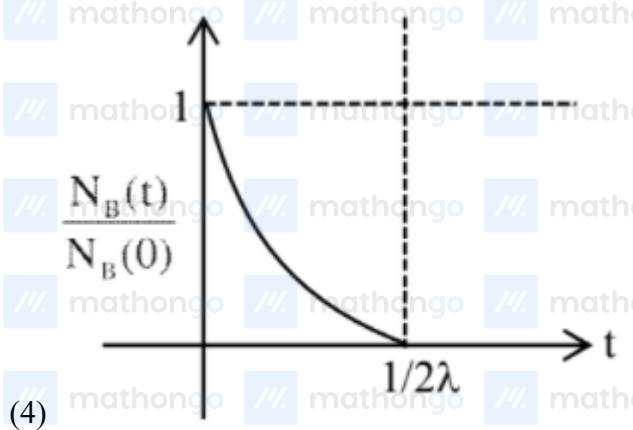
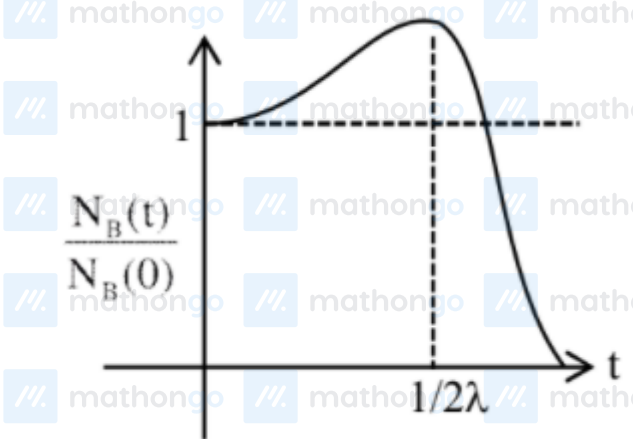
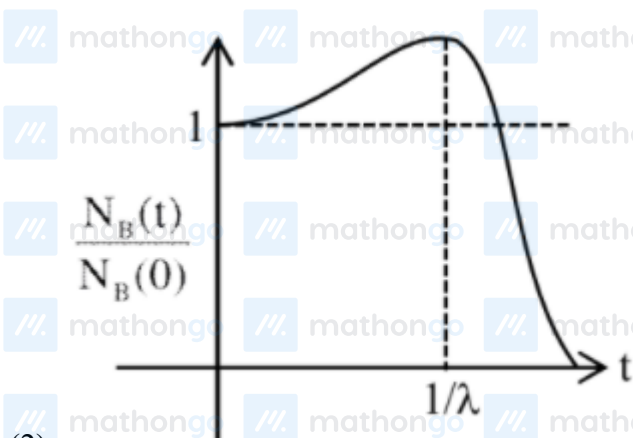
(4) 4×10^{10}

Q4 2021 (26 Aug Shift 2)

At time $t = 0$, a material is composed of two radioactive atoms A and B, where $N_A(0) = 2 N_B(0)$. The decay constant of both kind of radioactive atoms is λ . However, A disintegrates to B and B disintegrates to C. Which of the following figures represents the evolution of $N_B(t)/N_B(0)$ with respect to time t ?

$$\left[\begin{array}{l} N_A(0) = \text{No. of A atoms at } t = 0 \\ N_B(0) = \text{No. of B atoms at } t = 0 \end{array} \right]$$





Q4 (3)

Q1 (2)

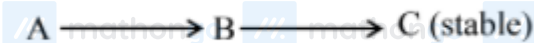
$$(t_{1/2})_x = (\tau)_y$$

$$\Rightarrow \frac{\ln 2}{\lambda_x} = \frac{1}{\lambda_y} \Rightarrow \lambda_x = 0.693 \lambda_y$$

Also initially $N_x = N_y = N_0$ Activity $A = \lambda N$ As $\lambda_x < \lambda_y \Rightarrow A_x < A_y$ $\Rightarrow$ y will decay faster than x

Option (2)

Q2 (2)



Initially no. of atoms of B = 0 after $t = 0$, no. atoms of B will start increasing & reaches maximum value when rate of decay of B = rate of formation of B.

After that maximum value, no. of atoms will start decreasing as growth & decay both are exponential functions, so best possible graph is (2)

Option (2)

Q3 (2)

$$\frac{N}{N_0} = \left(\frac{1}{2}\right)^{\frac{1}{412}}$$

$$\frac{N}{10^{10}} = \left(\frac{1}{2}\right)^{\frac{30}{60}}$$

$$N = 10^{10} \times \left(\frac{1}{2}\right)^{\frac{1}{2}} = \frac{10^{10}}{\sqrt{2}} \approx 7 \times 10^9$$

Q4 (3)



$$\frac{dN_B}{dt} = \lambda N_A - \lambda N_B$$

$$\frac{dN_B}{dt} = 2\lambda N_{B_0} e^{-\lambda t} - \lambda N_B$$

$$e^{-\lambda t} \left(\frac{dN_B}{dt} + \lambda N_B \right) = 2\lambda N_{B_0} e^{-\lambda t} \times e^{\lambda t}$$

$$\frac{d}{dt} (N_B e^{\lambda t}) = 2\lambda N_{B_0}, \text{ on integrating}$$

$$N_B e^{\lambda t} = 2\lambda t N_{B_0} + N_{B_0}$$

$$N_B = N_{B_0} [1 + 2\lambda t] e^{-\lambda t}$$

$$\frac{dN_B}{dt} = 0 \text{ at } -\lambda [1 + 2\lambda t] e^{-\lambda t} + 2\lambda e^{-\lambda t} = 0$$

$$N_{R_m} \text{ at } t = \frac{1}{2\lambda}$$